

24 March 2026

PRE-CLINICAL TESTING GUIDELINES AND CLINICAL REQUIREMENTS FOR PHASE I / FIRST IN HUMAN (FIH) STUDIES

This document has been prepared to serve as a guideline for the preparation of applications to conduct first-in-human (FIH) and Phase I clinical trials in South Africa. It is not intended as an exclusive approach, and SAHPRA reserves the right to request any additional information and may make amendments in keeping with the knowledge that is current at the time.

Document History

Final Version	Reason for Amendment	Effective Date [dd Month yyyy]
1	This is a new guideline for implementation	01 June 2026
2		

DR BOITUMELO SEMETE-MAKOKOTLELA
CHIEF EXECUTIVE OFFICER

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Glossary

Abbreviation / Key concept	Meaning
ABPI	Association of the British Pharmaceutical Industry guideline for insurance in clinical trials
AUC 0-tau	Area under the curve limited to the end of the dosing interval
Clinical studies/trials	These are Research studies conducted with human volunteers testing interventions to prevent, diagnose or treat disease
CTF-1	Clinical Trial Application Form
C _{max}	Maximum plasma concentration
DLT	Dose limiting toxicity
FIH	First-in-human
IP	Investigational product (including medicines, devices and IVDs)
IVD	In Vitro Diagnostic
In vitro	“In glass” or in a laboratory
In silico	In a computer
In vivo	In an animal or human
ICU	Intensive Care Unit
MAD	Multiple Ascending Dose
Medical Device (MD)	Means any instrument, apparatus, implement, machine, appliance, implant, reagent for in vitro use, software, material or other similar or related article, including Group III and IV Hazardous Substances contemplated in the Hazardous Substances Act, 1973 (Act No. 15 of 1973)— (a) intended by the manufacturer to be used, alone or in combination, for humans or animals, for one or more of the following: (i) diagnosis, prevention, monitoring, treatment or alleviation of disease; (ii) diagnosis, monitoring, treatment, alleviation of or compensation for an injury; (iii) investigation, replacement, modification or support of the anatomy or of a physiological process;

	(iv) supporting or sustaining life; (v) control of conception; (vi) disinfection of medical devices; or (vii) providing information for medical or diagnostic purposes by means of in vitro examination of specimens derived from the human body; and (b) which does not achieve its primary intended action by pharmacological, immunological or metabolic means, in or on the human or animal body, but which may be assisted in its intended function by such means
MTD	Maximum Tolerated Dose
NHREC	National Health Research Ethics Council (NHREC)
NOAEL	No-Observed-Adverse-Effect-Level in animal model
Non-clinical	All studies performed outside of humans, including pharmacology, pharmacokinetics, toxicology, safety pharmacology, to provide evidence of safety, guide dose selection and study design for clinical trials. These research activities may be conducted at any stage of product development.
PD	Pharmacodynamic: Study of the effects of a medical product in the human body. Pre-clinical PD studies can be used to show the effects in an animal model before starting human clinical trials.
PK	Pharmacokinetic: Studies to how the medicine levels vary over time depending on the way it is absorbed, distributed, how it is metabolised, and how it is excreted. Pre-clinical PK studies can be used to show these effects in an animal model before starting human clinical trials.
PBPK	Physiologically based Pharmacokinetic modelling is an experimental computational tool that simulates how a substance (medicine or chemical) moves through the body. ¹
Pre-clinical	Pre-clinical studies are non-clinical studies that are conducted before starting of human studies. Research activities done in lab / <i>in vivo</i> and <i>in vitro</i> prior to initiation of human studies. Laboratory research and animal studies are done to determine whether the experimental product has an acceptable safety profile before it is administered to humans.
SAD	Single Ascending Dose
SUSAR	Suspected Unexpected Serious Adverse Reactions

¹ D Deepika et al., Int J Environ Res Public Health. 2023 Feb 16;20(4):3473.

1. INTRODUCTION

Pre-clinical Testing

Information from pre-clinical testing is essential in planning clinical trials in humans. For first-in-human (FIH) trials, the information is used to decide what the starting dose should be, and the range of doses to be tested. It also suggests what clinical signs and symptoms as well as laboratory abnormalities should be looked for to detect any adverse effects, especially Serious Adverse Events (SAEs).

This document is intended to guide the type and extent of pre-clinical data to be provided in the Clinical Trial Application Form (CTF-1) for submission to SAHPRA for the conduct of a clinical trial in South Africa. Detailed data must be provided in the Investigator's Brochure and cross-referenced from the CTF-1 where appropriate. In the CTF-1, the applicant is required to provide a summary of the pre-clinical testing that has been conducted during the development of a candidate medicinal or relevant related health products, and a discussion of how this supports the design of the clinical trial in the application. Pre-clinical testing may include both *in vitro* (laboratory), *in silico* (computer) and *in vivo* (animal) studies.

The Guideline is adopted from ICH M3 (R2) non-clinical safety studies for the conduct of human clinical trials for pharmaceuticals - Scientific guideline².

SAHPRA supports the current international principles of the 3Rs - Reduction, Refinement, and Replacement – in animal testing, which involves animal pain, suffering, and/or death. However, in the development of new medicines certain animal tests remain essential for ensuring human safety before clinical investigation.

Pre-clinical tests results supporting product functions and safety profile are particularly relevant to support FIH clinical trials but are also important support for subsequent and more extensive clinical trials where physiological or adverse effects observed in animal models can be important for the design of safety surveillance during later phases of clinical development.

Pre-clinical test results can provide a preliminary assessment of the balance of risks versus potential benefits of the candidate investigational product (IP).

² [<https://www.ema.europa.eu/en/ich-m3-r2-non-clinical-safety-studies-conduct-human-clinical-trials-pharmaceuticals-scientific-guideline>]

Animal testing can be used for elucidation of biodistribution, metabolism and elimination and evaluation of the investigational product activity if a suitable animal model is available. Animal models should be validated and appropriate for the class of medicine/device being tested.

Small-animal models, including mice and rats, may be useful for toxicity testing, but larger animal species are more appropriate for investigating physiological functions and may include rabbits, dogs, sheep, pigs and non-human primates.

Toxicity testing of pharmaceutical medicines should establish a No-Observed-Adverse-Effect-Level (NOAEL) in an appropriate animal model and may be determined during repeat dose toxicity studies. Pre-clinical testing of medical devices is necessary to establish safety and functional capabilities. Where animal models are needed, they must be qualified as appropriate for establishment of the required endpoint that maybe anatomical or physiological.

In addition, for Biosimilar medicines, the investigational product must be fully characterised by comparative physico-chemical testing, demonstration of functional activity (e.g., receptor binding in cell cultures), and *in vivo* testing as “Biosimilar³” before it can be included in clinical trials. SAHPRA may consider that for biosimilar medicines, established by comparative physico-chemical and functional similarity studies, in-vivo testing may be unnecessary.

Clinical Factors in early Clinical Trials

The Clinical factors section outlines the key clinical considerations relevant to dose selection, dose escalation, maximum exposure, and transition between single and multiple dosing regimens, with particular emphasis on healthy volunteer studies. It also addresses the planning and conduct of early clinical trials, including protocol design, participant selection, safety monitoring, data review, stopping rules, and communication of adverse events. In addition, the responsibilities of sponsors and investigators, as well as requirements for clinical site facilities and emergency preparedness, are described. Collectively, these principles aim to support informed decision-making, minimize risk, and ensure regulatory compliance throughout the early stages of clinical development.

2. PURPOSE AND SCOPE

³ SAHPGL-PEM-BIO-02_v4 Biosimilar medicine Q S E _ Final version for BMEAC 06 Dec 2024

This guideline sets out the key requirements and safeguards for pre-clinical testing and for First-in-Human (FIH)/ Phase I clinical studies in South Africa.

This document applies to the situations usually encountered during the development of pharmaceuticals, in case of studies involving medical devices/ IVDs (e.g., diagnostic assays and kits, monitoring systems), refer also to the Guidelines on clinical investigation requirements on <https://www.sahpra.org.za/medical-devices-and-in-vitro-diagnostics-guidelines>.

3. LEGAL PROVISION

All clinical studies must comply with national regulations and international best practice. This includes compliance with but not limited to:

- Applicable provisions of the Medicines and Related Substances Act 101 of 1965 (as amended) and its Regulations.
- The National Health Act 61 of 2003 and regulations governing health research involving human participants.
- National Health Research Ethics Council (2024) South African Ethics in Health Research Guidelines: Principles, Processes and Structures, 3rd ed. National Department of Health of the Republic of South Africa.
- South African Good Clinical Practice (SAGCP) guidelines adopted by SAHPRA

In alignment with the latest:

- Declaration of Helsinki
- ICH E6 standards on Good Clinical Practice
- ICH M3 (R2): Guideline on pre-clinical safety studies for the conduct of human clinical trials
- CIOMS guidelines

4. BACKGROUND

The clinical development of a medicine follows a sequential series of steps, from drug-discovery through pre-clinical testing to clinical trials and eventual marketing authorization (Regulatory approval) (Registration/Marketing approval/Licence). The results from each stage must support progression to the subsequent stage.

Pre-clinical		Phase-I		Phase-II	Phase-III	Phase-IV
In-vitro <i>/in-silico</i>	In-vivo	1 st in human Dose Finding and Safety	Special groups including healthy volunteers	Proof of concept, Safety & preliminary efficacy	Safety & Efficacy (sufficient powering and randomisation)	Post approval Safety in use, especially long-term AEs/SAEs

Pre-clinical testing is required to establish the safety profile, pharmacology, toxicology, and pharmacokinetics of the IP before clinical (in humans) testing can be initiated.

For Phase-II and Phase-III trials, results from pre-clinical including non-clinical testing can guide surveillance for potential rare side effects.

The goals of the pre-clinical safety assessment programme include establishing the initial safety profile, including the characterisation of adverse effects, especially serious Adverse Events, the identification of target organs, clarification of dose dependence, the relationship of toxicities to exposure, and potential reversibility. Tests of the functionality and metabolism of the candidate molecule (primary pharmacodynamics and Pharmacokinetics) can be explored *in vitro* and in animal models. These characteristics can be confirmed for experimental formulations of the candidate medicine in the same systems.

The following pre-clinical studies should be reported (based on ICH M3) to support FIH studies:

- Pharmacology studies,
- General toxicity studies,
- Toxicokinetic and pre-clinical pharmacokinetic studies, and
- Repeated-dose toxicity studies.

Further non-clinical studies will be conducted on a case-by-case basis and at any phase, including:

- o Assessments of carcinogenic potential
- o Phototoxicity, immunotoxicity, juvenile animal toxicity, etc.
- o For biotechnology-derived products (see ICH topic S6)
- o For life-threatening or serious diseases – such as resistant infections, cancer, etc., that do not have a current adequately effective therapy.
- o Teratogenicity: Teratogenicity, Developmental and Reproductive toxicity studies will be required before exposure of women of child-bearing potential to the investigational drug or comparators.

- o Platform technologies: Where a medicine or vaccine are of the same or similar class and are based on a well-accepted formulation with minor modifications to the active ingredient, then the applicant can request that the pre-clinical program be reduced. This is particularly relevant in biosimilar studies.

Where animal testing is required, examinations for adverse effects and toxicity need to be conducted in suitable animal models and in more than one family of animals (not all rodents or non-human primates), at least one of which is considered relevant to the human condition. Conduct of research in animals in South Africa requires approval from the institutional Animal Ethics Committee.

5. DETAILS OF PRE-CLINICAL TESTS THAT NEED TO BE REPORTED

Safety pharmacology core studies

Assessing effects on the cardiovascular, respiratory, renal, gastrointestinal, haematological and central nervous systems

Primary pharmacodynamics studies

In vivo and/or *in vitro* studies, assessing the mode of action/effects of the candidate compound on the target.

Pharmacokinetics and toxicokinetics studies

Data gathered during *in vitro* studies on metabolic and blood protein binding data for animals and humans.

Systemic exposure data from toxicology studies. Biodistribution and elimination.

These studies are not usually required for biological medicines apart from immunotherapeutic medicines, blood products, live vaccines, or gene products.

Acute toxicity studies

Single-dose toxicity studies in more than one mammalian species can be completed during studies that define dose limiting toxicity (DLT) and maximum tolerated dose (MTD) in the species used for toxicity testing. The NOAEL relevant to the intended use of the IP can be accepted as a guide to estimating the maximum dose to be used in clinical trials.

Repeated-dose toxicity studies

Vary in length according to duration, therapeutic indication, and scope of the proposed clinical program.

Minimum duration for two weeks in two mammalian species (one of which is not a rodent) but may need to be extended to be relevant for the proposed trial duration.

Repeat-dose toxicity studies can be designed to collect data supporting single-dose acute toxicity studies.

Reversibility of toxicity

Animal studies can indicate the potential for reversibility of adverse events that are detected.

Other studies of concern

e.g., investigation of phototoxicity and/or radiation toxicity

Teratogenicity: Pregnant women must be excluded in all first in human studies. Preclinical and clinical results from teratogenicity, developmental and reproductive toxicity studies will be required before exposure of women of child-bearing potential to any investigational drug or comparators.

In the extremely rare case that the participant falls pregnant while on FIH/phase I study, the participant should be terminated immediately from study treatment and the pregnancy and the baby be carefully followed up. Please refer to the SAHPRA Guidelines on Pregnancy and Lactation.

6. KEY POINTS FOR CTC EVALUATION OF PRE-CLINICAL DATA TO SUPPORT PHASE I CLINICAL TRIAL

- Have the relevant non-clinical tests as required by ICH and other guidelines been conducted and reported in this application? Is there an acceptable justification for any omissions?
- Is there adequate evidence of safety and evidence supporting efficacy from pre-clinical studies to proceed with clinical studies, particularly for early phase or FIH clinical trials? Is there an acceptable justification for any omissions?
- For the FIH clinical study, are the proposed starting dose, dose escalation, and maximum exposure justified based on the pre-clinical data for this application?
- Has the current formulation been included in the reported pre-clinical test results? Are the pre-clinical data relevant to the proposed formulation and route of administration in the clinical study application?
- Where medicines are used in combination, special consideration needs to be given to the potential adverse effects and the dosages used of both the IP and the registered medicine(s).
- Are there additional surveillance activities for participants that should be conducted during the proposed clinical trial, based on the reported results from the pre-clinical studies (or potential adverse effects reported from medicines in the same class)?
- Development and Reproductive Toxicity (DART) and teratogenicity studies are not usually required for FIH studies, as the safety of the candidate IP given to non-pregnant women must be established before trials in pregnant or breastfeeding women can be undertaken.

7. CLINICAL FACTORS: CONSIDERATIONS FOR PHASE-I/FIH CLINICAL TRIALS

7.1 Dose selection for FIH and early clinical trials

The starting dose and maximum exposure, as well as dose escalation steps during the clinical trial, should be justified and outlined in the protocol. The methods used and calculations on how doses and estimated

exposure levels were determined, including methods for modelling (e.g., PK/PD and physiologically based pharmacokinetic (PBPK)), should be summarised in the CTF-1 and protocol and the details included in the investigational brochure (IB).

Deviations from the prespecified dose escalation and decision-making criteria would warrant the submission of substantial amendment(s). Decision making criteria for adapting dosage based on clinical data based on emerging clinical data should be described.

In addition, sentinel dosing of two participants (one active, one placebo) is required at a minimum for the first single ascending dose (SAD) and multiple ascending dose (MAD) cohorts.

7.1.1 Starting dose for Healthy Volunteers

Sentinel safety cohorts are small groups of participants dosed before the main cohort. These may help to identify potential adverse effects, and mitigate risk, especially in FIH and Phase 1 trials.

The starting dose must mitigate the risk of toxicity while balancing the need to elicit pharmacologic activity. Reference should be made to published guidance documents from EMA and the FDA for investigators to follow when determining an appropriate starting dose.

There are two methods that may be used for calculating the starting dose; an empirical or a mechanistic approach.

The empirical approach aims at avoiding toxicity at initial human doses by using the NOAEL, a generally accepted benchmark for safety (FDA). The exposures achieved at the NOAEL in the most relevant animal species used (which might not necessarily be the most sensitive species) should be used for the estimation of an equivalent exposure for humans.

The mechanistic approach is based on state-of-the-art modelling incorporating all relevant preclinical pharmacology data, including *ex vivo* and *in vitro* studies in human tissues (EMA).

This approach enables the selection of a minimal anticipated biological effect dose-level, as this is becoming increasingly important as more targeted therapies are being developed that are very specific and potent, with predicted pharmacologically active doses well below a dose that is thought to be safe based on the NOAEL.

7.1.2 Maximum Dose Selection

The maximum exposure should be justified based on all available pre-clinical and clinical data, including PD, PK, findings in toxicity studies, and exposure at the expected therapeutic dose range. An expected maximum exposure level must not be exceeded in the study without approval of a substantial protocol amendment, this should have been pre-defined in the protocol for each study part.

7.1.3 Moving from single and multiple dosing

The selection of an appropriate dosing interval and duration of dosing for all multiple dosing cohorts and study parts should consider the specific PK and PD characteristics of the IP, the available non-clinical safety data, and all data from participants in previous single-dose cohorts.

A maximum duration of dosing should be stated in the protocol for every cohort. The expected exposure after multiple dosing (C_{max} and $AUC_{0-\tau}$) should have been covered during preceding SAD parts/trials.

7.1.4 Route of Administration

The choice of route of administration for dosing in humans should be based on the pre-clinical data, the characteristics of the IP, and the intended therapeutic use.

7.2 Planning and conducting early clinical trials

7.2.1 Protocol - Overall design

The following are key considerations:

- a) Description of the trial and its ethical and scientific rationale. This would include a description of what is being done in the trial and the rationale behind key decisions so that the trial can be subject to scrutiny in regulatory assessment.
- b) Graphical representation of the overall scheme of the proposed trial in real-time, showing intervals for interventions, sampling, timing of all reviews, and decision points.
- c) Details on the size of the cohorts, including how many participants are on active IP and how many are on placebo treatment.
- d) Phase-1 clinical trials typically enroll small groups of participants; about 20 per cohort, and less than 100 for the whole trial.

7.2.2 Integrated protocol

The practice of conducting FIH/early clinical trials with integrated protocols means that the information generated in previous parts needs to be analysed and integrated into an assessment in a limited timeframe before deciding on proceeding to the next part.

All parts, and the criteria to move from one part to another, should be predefined within an integrated protocol, as should possible modifications, based on the totality of available information and the related uncertainty.

Recommendations for time sequence for the conduct of different parts:

- a) Overlapping of SAD and MAD parts may be acceptable. The proposed overlap, however, should be scientifically justified and supported by decision points and a review of available data before starting the MAD part.

- b) Other single-dose parts (e.g., food interaction) could be conducted in parallel to the SAD part, provided the dose chosen and the expected exposure are equal to or lower than that which was reached in a concluded preceding SAD cohort where all relevant data have been reviewed and no dose escalation stopping criteria were met.
- c) Other study parts that involve multiple dosing (e.g., drug-drug interaction) should generally not overlap with earlier SAD or MAD cohorts. All relevant SAD and MAD data should be reviewed before starting these parts. Deviation from this should be justified in the protocol.

7.2.3 Choice of participants

The choice to conduct FIH trials in healthy participants or patients should be made on a case-by-case basis. The risks of enrolling healthy volunteers versus patients must be weighed, including toxicities, PK variability, lifestyle conditions, benefits to patients, and special populations.

Clinical factors to consider in the decision to conduct a study in healthy volunteers or patients include:

- a) Whether the toxicities foreseen/risks associated can support the inclusion of healthy volunteers.
- b) The relative presence of the molecular target in healthy participants or in patients.
- c) The possible variable PK, PD or safety profile in participants.
- d) The potential differences between the targeted patient group and healthy volunteers.
- e) Possible interactions with participants' lifestyle, e.g., smoking, use of alcohol or recreational drugs.
- f) The use of other medications with the possibility of adverse reactions and/or difficulties in the interpretation of results.
- g) A potential of patients to benefit from other products or interventions.
- h) The predicted therapeutic window of the IP based on preclinical testing.
- i) Factors relating to special populations, including age, gender, ethnicity, and genotype(s).

Key inclusion and exclusion criteria for trials involving healthy participants should include an adequate set of vital signs (including ECG), laboratory values and clinical assessments that should be within normal ranges. A justification for deviations outside the defined normal ranges should be provided.

7.2.4 Safety monitoring and interventions

The detection of a safety signal with a high degree of certainty is extremely limited by the typically small size of these trials. Successful signal detection in FIH studies depends on adequate collection of data coupled to a systematic analysis of the data, a determination of the strength of a potential signal, and an understanding of the quantitative limitations of typical study designs.

The participant's safety assessments that will be routinely conducted, their timing, and any additional monitoring actions or interventions should be pre-specified in line with the known pharmacological and non-

clinical safety profile and balanced against the degree of uncertainty. Any proposal to routinely omit an assessment should be scientifically based.

The length of follow-up of participants should be specified within the protocol (e.g., for possible delayed adverse reactions).

The sponsor should describe how safety monitoring should be extended until parameters return to within the normal range or to baseline, as appropriate for the population.

Extended monitoring should be considered, e.g., when the mechanism entails enzyme inhibition or activation (monitoring should continue until enzyme activity has returned to baseline or to an acceptable percentage of baseline) or when prolonged PD effects are observed, regardless of duration of target inhibition or the PK profile of the IP.

Repeated assessments, integrating available knowledge with rapid processing of emerging information, are crucial for the recognition of developing toxicity at an early stage.

7.2.5 Data review for decisions

The data supporting dose escalation or the beginning of a new study part in alignment with the predefined criteria in the protocol are key and should be described in the protocol.

The following are regarded as minimum criteria for data review:

- Evaluation of data by a defined expert group and summary of criteria for evaluation and progression to the next dose-level.
- ‘Evaluable’ participants should be defined, i.e., participants who have completed all planned study visits at least until the time of the decision as detailed in the protocol.
- Data collection as planned in the protocol in each dosing cohort should be complete and the decision to proceed to the next dose cohort should be documented.

7.2.6 Early Stopping rules

The protocol should define unambiguous stopping rules that result in an immediate stop to dosing.

Stopping rules should be defined for each of the following:

- Final stop to dosing and termination of the trial;
- Stopping for an individual participant, at any time in the trial;
- Stopping within a cohort
 - when participants in a cohort are dosed staggered;
 - during multiple dosing;
- Progression to the next part of the trial;

- Any dose escalation parts of the trial.

Stopping rules with healthy volunteers must include precautions in the event of the following:

- a 'serious adverse reaction, considered at least possibly related to the IP administration, in one participant;
- 'severe' non-serious adverse reactions, possibly related to the IP administration, in two participants in the same cohort, independent of within or not within the same system-organ-class.

7.2.7 Monitoring and communication of adverse events

For all clinical trials, adverse events of special interest (AESI) should be defined, based on non-clinical findings and observations from experience with other similar active ingredients. Suitable monitoring activities for these AESIs should be described.

Due to high risk of Phase 1 /FIH trials, four weekly progress reports on a SAHPRA template should be submitted.

7.3 Documentation of sponsor and investigational responsibilities

The responsibilities of the sponsor and investigator(s) (as well as any other experts or study staff) in decision-making should be clearly defined in the protocol. Responsibility regarding breaking the treatment code in emergencies should also be documented.

The composition of any decision-making group or safety review committee should be documented in the protocol. Other details to include are the exact remit of the group and the roles of all members in the committee, and their relation to the sponsor. Consideration should be given to the inclusion of independent experts who are (at least) external to the study. Written statements and conclusions by any decision-making or safety review group must be in place before allowing trial progression at the noted times as per protocol. This includes documentation of appropriate quality control checks on the data reviewed.

7.3.1 Investigator site facilities/personnel

To prioritize participant safety and data integrity, FIH clinical trials should be undertaken in appropriate clinical facilities and conducted by investigators and medical staff with the appropriate levels of training and experience in early-phase clinical trials. Relevant medical expertise and up-to-date GCP training, as well as an understanding of the special characteristics of the IP and its mechanism of action are crucial.

It is a necessity for FIH clinical trials to take place under controlled conditions, such as inpatient care, where close supervision of trial participants during and after dosing can be provided. Sites must have access to emergency supplies, and staff must be prepared for medical emergencies.

It is advisable that FIH clinical trials should be conducted at a single site. Where multiple sites are required for enrolment, protocols should be in place providing a description of measures to reduce risks that might occur from the use of multiple sites.

Procedures should be established between the clinical research unit and its nearby intensive care unit (ICU) regarding the responsibilities and undertakings of each in the transfer and care of patients.

7.3.2 Referrals to hospital, including Intensive Care Unit (ICU)

Participants in all clinical studies, especially Phase I / FIH studies, who develop serious adverse event(s) requiring hospitalization need to be admitted to a nearby hospital with facilities to deal with such urgent problems. This includes access to imaging and interventional radiology, laboratory services, pharmacy, emergency admission and resuscitation, as well as intensive care (ICU). ICU facilities should cover the appropriate SAE, and may need to be Medical, Surgical, or Paediatric ICUs. The research site / PI must have prior arrangements with the appropriate specialist(s) at the nearby hospital to enable rapid admission and avoid long ER/casualty waiting. Based on ABPI, the Sponsor/Applicant must cover the costs of such admissions and facilitate immediate payments to the applicable hospital.

8. FINANCIAL FACTORS

8.1 Participant remuneration:

Please refer to the SAHPRA Guideline for Remuneration of Volunteers in Phase I Clinical Trials.

8.2 Sponsor responsibility:

In the case of interventional (including Phase I) trials, the sponsor must provide comprehensive no-fault insurance/liability cover for trial-related injury or death and must indemnify investigators and trial sites for liabilities arising from trial conduct (except where investigator negligence/malpractice is proven).

8.2.1 No-fault insurance/liability cover:

On submission to SAHPRA, an up-to-date insurance certificate must be included that contains:

- Insurer name, local address & contact person/phone.
- Policy number insured party (sponsor), a clear statement that it covers trial-related bodily injury, including death.
- Scope of cover (no-fault compensation for medical treatment for serious/enduring injury and death compensation to the family).
- Dates: cover inclusive of approval date, trial procedures, and a reasonable period for late injuries.
- Any special conditions (should not invalidate/abate cover); no deductibles or co-payments charged to participants.
- Declaration of compliance with SA GCP (and reference to ABPI approach).
- Policies must be underwritten in line with the SA Insurance Act. All local insurer/broker details must be provided to facilitate easy access.

Importantly, insurance must not impose deductibles/co-payments that the participant must pay. If insurance policies have deductibles, they must be covered fully by the Sponsor/Applicant. Cover must include a reasonable period for late-onset injuries. This should be stated clearly in the informed consent form.

8.2.2 Indemnification of investigators and trial site:

The indemnification of investigators and trial sites for liabilities arising from trial conduct (except where investigator negligence is proven) is mandatory and should be indicated in the clinical trial agreement with the investigator.

8.2.3 Investigator negligence/malpractice (professional indemnity)

The malpractice cover is separate from sponsor trial insurance and site indemnification cover. All healthcare professionals involved in all clinical trials must maintain personal professional indemnity/malpractice cover to cover clinical negligence or investigator malpractice. Trial insurance does not replace the clinicians' own indemnity.

9. FACILITY REQUIREMENTS FOR PHASE I CLINICAL TRIALS UNITS:

9.1 Physical layout and participant accommodation

- Dedicated inpatient observation area or ward (beds for overnight monitoring of healthy volunteers) with a visible nursing station and monitoring capability. Phase I studies often requires prolonged observation, and the extent of this requirement will be dependent on the design of the protocol.

9.2 Emergency and resuscitation capability

- Resuscitation trolley/crash cart stocked as per local emergency SOP and in keeping with SAHPRA Guidelines (Emergency Procedures for Clinical Trial Sites).
- Written emergency procedures (including rapid unblinding and emergency referrals) and contact list for nearby emergency services/hospital transfer plan.

9.3 Medical equipment & monitoring

All medical equipment needs regular and current calibration. Any specialised emergency equipment and medicines required for a specific trial must be listed.

10. ETHICS AND REGULATORY FACTORS:

10.1 Role of RECS and the NHREC:

Due to their inherently high-risk nature, Phase I studies, particularly first-in-human trials (FIHs), also require heightened attention from Health Research Ethics Committees (RECs).

In most phase 1 / FIH studies with healthy volunteer participants, reviewers need to balance the potential risk of harm with possible future benefits to the general community, while in the case of oncology and gene therapy studies, reviewers need to balance the risk of harm with any potential benefits to the both the participant and to the general community.

Special attention should be given to preclinical data, dose selection and escalation, safety monitoring plans, stopping rules, participant selection criteria, and the adequacy of informed consent processes. In the case of oncology and gene therapy studies, additional attention should be given to the underlying disease process and comorbidities of the study participants.

In addition, when reviewing such studies, the REC reviewers must particularly follow both the latest SAGCP Guidelines and the National Department of Health (NDoH) Ethics in Research Guidelines, developed by the National Health Research Ethics Council (NHREC).

Through this structured oversight framework, RECs and the NHREC play a critical role in ensuring that early-phase clinical research in South Africa is conducted ethically, responsibly, and in a manner that protects participant safety while advancing scientific knowledge.

10.2 Other Regulatory Authorities:

When reviewing Phase I and especially FIHs in South Africa, information used and decisions made, particularly concerning the risk of harm, made by other Regulatory Authorities that SAHPRA aligns with, e.g., FDA, EMA, MHRA, when available, should be considered. The decisions and policies from the African Medicines Agency (AMA) and Zambia, Zimbabwe, Botswana, Namibia, and South Africa (ZAZIBONA collaboration) should be considered concerning FIH / Phase I clinical trials.

10.3 Reporting frequency

As these studies are considered high risk, or indeed very high risk, more frequent reporting to both SAHPRA and the applicable RECs is needed.

All FIHs require four weekly routine reports as well as SAE reporting within 24 hours, as is usually the case. Phase I clinical trials other than FIH may also require more frequent reports as per the potential risk of harm and as determined by SAHPRA/CTC.

10.4 Audits and inspections

As FIHs require highly trained staff as well as special facilities where participants may be managed and possibly referred, in the case of an SAE, a pre-study inspection by the SAHPRA GCP Inspectorate may be required. Thereafter, both routine inspections and unannounced inspections may occur. Adequate funding must be provided by the Sponsor/Applicant for the research site to allow for such monitoring activities.

11. REFERENCES

The following related documents are referenced:

1. WHO TRS 927 Handbook for Non-Clinical Safety Testing: <https://www.who.int/teams/health-product-policy-and-standards/standards-and-specifications/norms-and-standards/vaccine-standardization/non-clinical-evaluation-of-vaccines>.
2. International Conference on Harmonisation (2009). Guidance on nonclinical safety studies for the conduct of human clinical trials and marketing authorisation for pharmaceuticals M3(R2). Step 4 Geneva: ICH. from US FDA 2010: <https://www.fda.gov/media/71542/download>
3. International Conference on Harmonisation (2011). Preclinical safety evaluation of biotechnology-derived pharmaceuticals S6(R1). Step 4 version. Geneva: ICH. Retrieved 27.7.2015, from: EMA https://www.ema.europa.eu/en/documents/scientific-guideline/ich-s6r1-preclinical-safety-evaluation-biotechnology-derived-pharmaceuticals-step-5_en.pdf

12. VALIDITY

This guideline is valid for a period of five (5) years from the date of publication. It will be reviewed periodically and may be updated earlier if new scientific, ethical, or regulatory information becomes available.